

```
In [49]: # Insight: Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [50]: # Insight: Load dataset from Excel file
fd = pd.read_excel("loan_data.xlsx")
fd
```

```
Out[50]:
```

	Applicant_ID	Income	Loan_Amount	Credit_Score	Employment_Type	Loan_Status
0	701	30000	200000	650	Salaried	Rejected
1	702	50000	300000	720	Self-Employed	Approved
2	703	40000	250000	680	Salaried	Approved
3	704	80000	500000	750	Salaried	Approved
4	705	60000	350000	700	Self-Employed	Approved
5	706	35000	220000	640	Salaried	Rejected
6	707	90000	600000	780	Salaried	Approved
7	708	45000	280000	690	Self-Employed	Approved
8	709	70000	450000	740	Salaried	Approved
9	710	55000	320000	710	Salaried	Approved

```
In [51]: # Insight: Display first 5 records
fd.head()
```

```
Out[51]:
```

	Applicant_ID	Income	Loan_Amount	Credit_Score	Employment_Type	Loan_Status
0	701	30000	200000	650	Salaried	Rejected
1	702	50000	300000	720	Self-Employed	Approved
2	703	40000	250000	680	Salaried	Approved
3	704	80000	500000	750	Salaried	Approved
4	705	60000	350000	700	Self-Employed	Approved

```
In [52]: # Insight: Display last 5 records
fd.tail()
```

Out[52]:

	Applicant_ID	Income	Loan_Amount	Credit_Score	Employment_Type	Loan_Status
5	706	35000	220000	640	Salaried	Rejected
6	707	90000	600000	780	Salaried	Approved
7	708	45000	280000	690	Self-Employed	Approved
8	709	70000	450000	740	Salaried	Approved
9	710	55000	320000	710	Salaried	Approved

In [53]: *# Insight: Check dataset information (data types, null values)*
`fd.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Applicant_ID    10 non-null    int64
1   Income          10 non-null    int64
2   Loan_Amount     10 non-null    int64
3   Credit_Score    10 non-null    int64
4   Employment_Type 10 non-null    object
5   Loan_Status     10 non-null    object
dtypes: int64(4), object(2)
memory usage: 612.0+ bytes
```

In [54]: `fd1 = {`
`"Income_Level":["Low","Medium","Medium","High","High","Low","High","Medium","High"`
`"Credit_Category":["Fair","Excellent","Good","Excellent","Good","Poor","Excellent"`
`}`

In [55]: *# Insight: Convert dictionary into DataFrame*
`fd1 = pd.DataFrame(fd1)`
`fd1`

Out[55]:

	Income_Level	Credit_Category
0	Low	Fair
1	Medium	Excellent
2	Medium	Good
3	High	Excellent
4	High	Good
5	Low	Poor
6	High	Excellent
7	Medium	Good
8	High	Excellent
9	Medium	Excellent

```
In [56]: # Insight: Merge original dataset with new dataset
Loan_Data = pd.merge(fd, fd1, left_index=True, right_index=True)
Loan_Data
```

```
Out[56]:
```

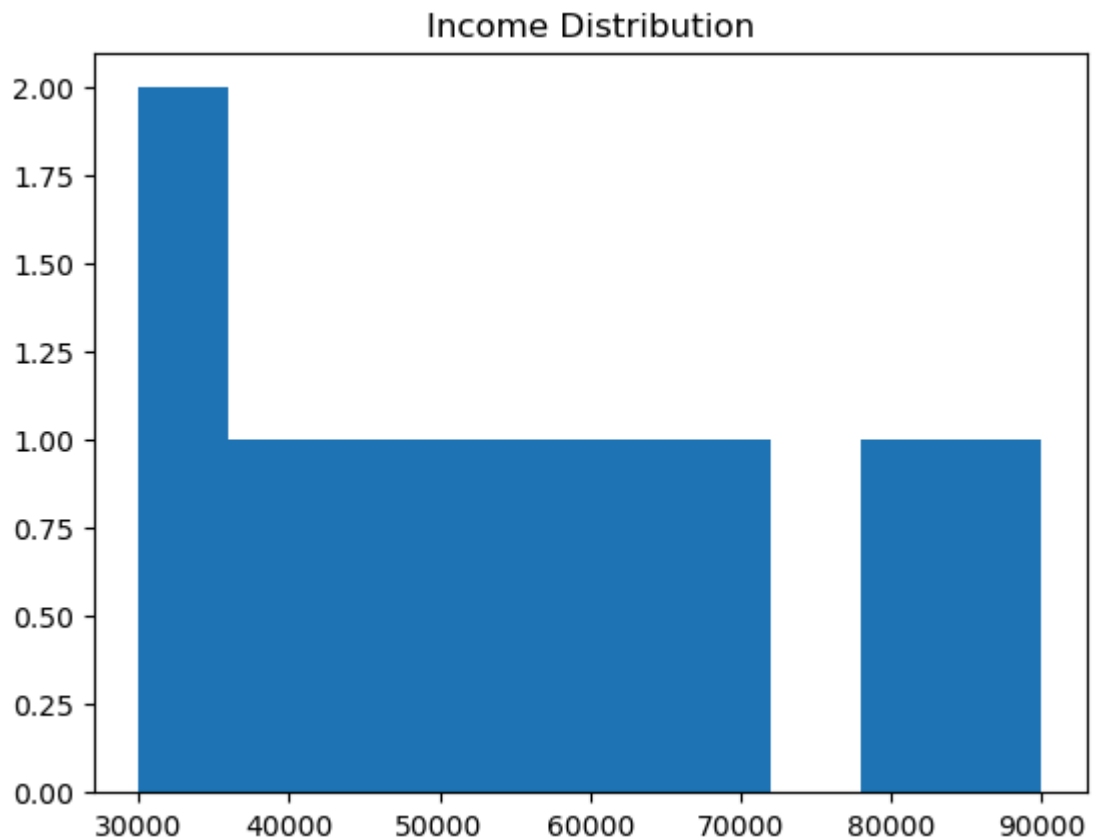
	Applicant_ID	Income	Loan_Amount	Credit_Score	Employment_Type	Loan_Status
0	701	30000	200000	650	Salaried	Rejected
1	702	50000	300000	720	Self-Employed	Approved
2	703	40000	250000	680	Salaried	Approved
3	704	80000	500000	750	Salaried	Approved
4	705	60000	350000	700	Self-Employed	Approved
5	706	35000	220000	640	Salaried	Rejected
6	707	90000	600000	780	Salaried	Approved
7	708	45000	280000	690	Self-Employed	Approved
8	709	70000	450000	740	Salaried	Approved
9	710	55000	320000	710	Salaried	Approved

```
In [57]: # Insight: Merge datasets
Loan_Data.groupby("Credit_Category")[["Income", "Loan_Amount"]].mean()
```

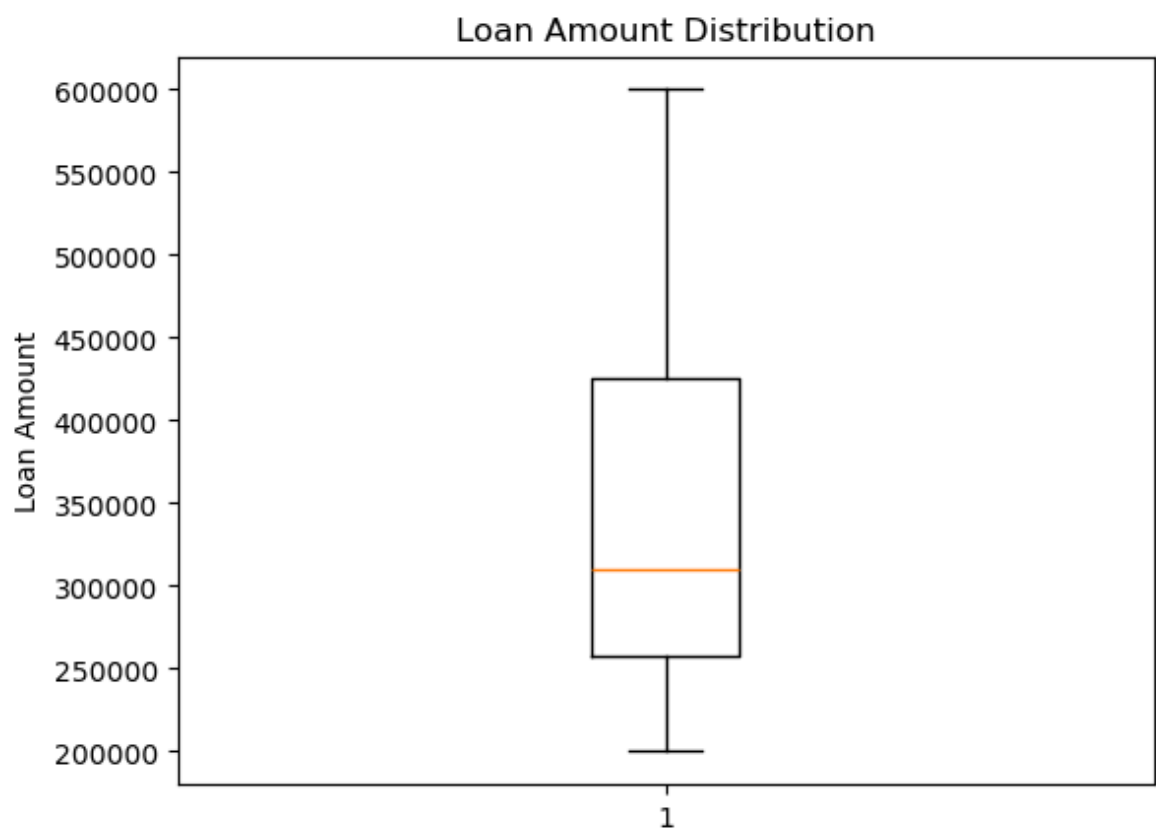
```
Out[57]:
```

	Income	Loan_Amount
Credit_Category		
Excellent	69000.000000	434000.000000
Fair	30000.000000	200000.000000
Good	48333.333333	293333.333333
Poor	35000.000000	220000.000000

```
In [70]: # Insight: Histogram
plt.hist(Loan_Data["Income"])
plt.title("Income Distribution")
plt.show()
```

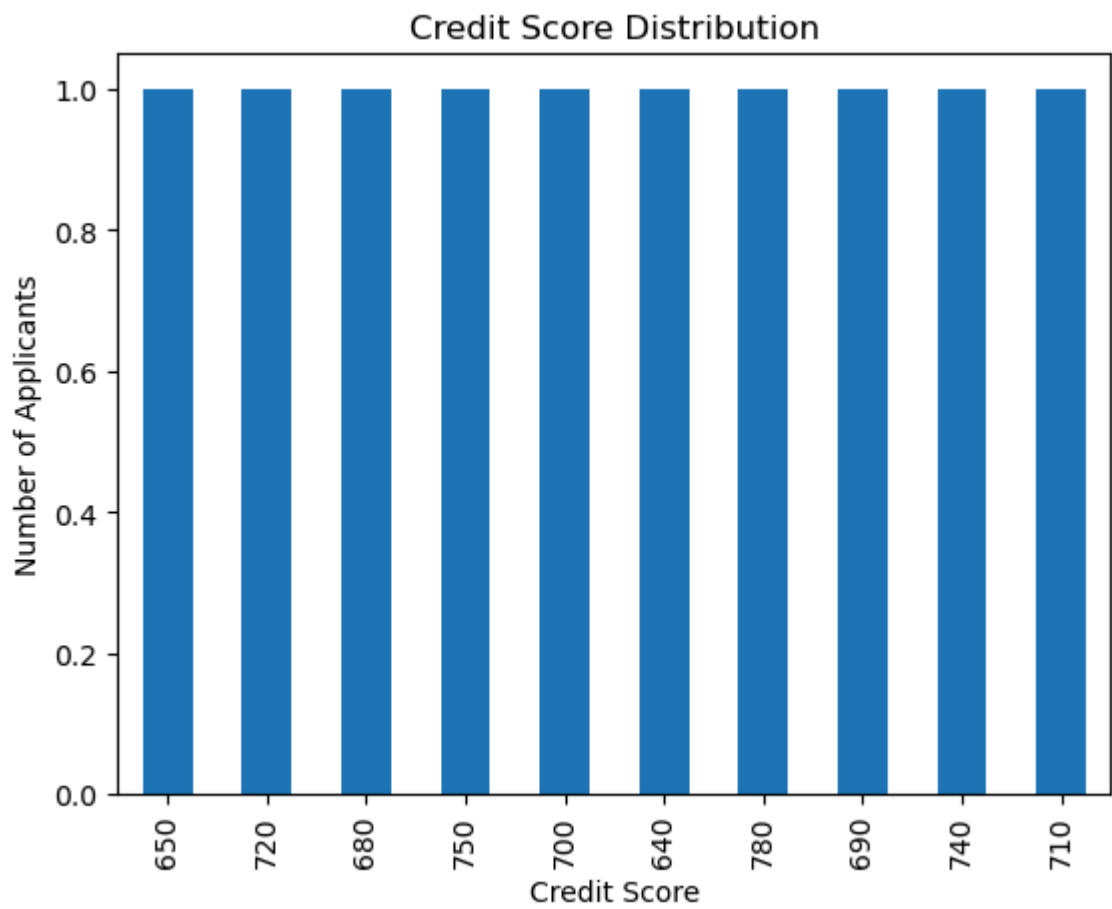


```
In [69]: # Insight: Boxplot
plt.boxplot(Loan_Data["Loan_Amount"])
plt.title("Loan Amount Distribution")
plt.ylabel("Loan Amount")
plt.show()
```

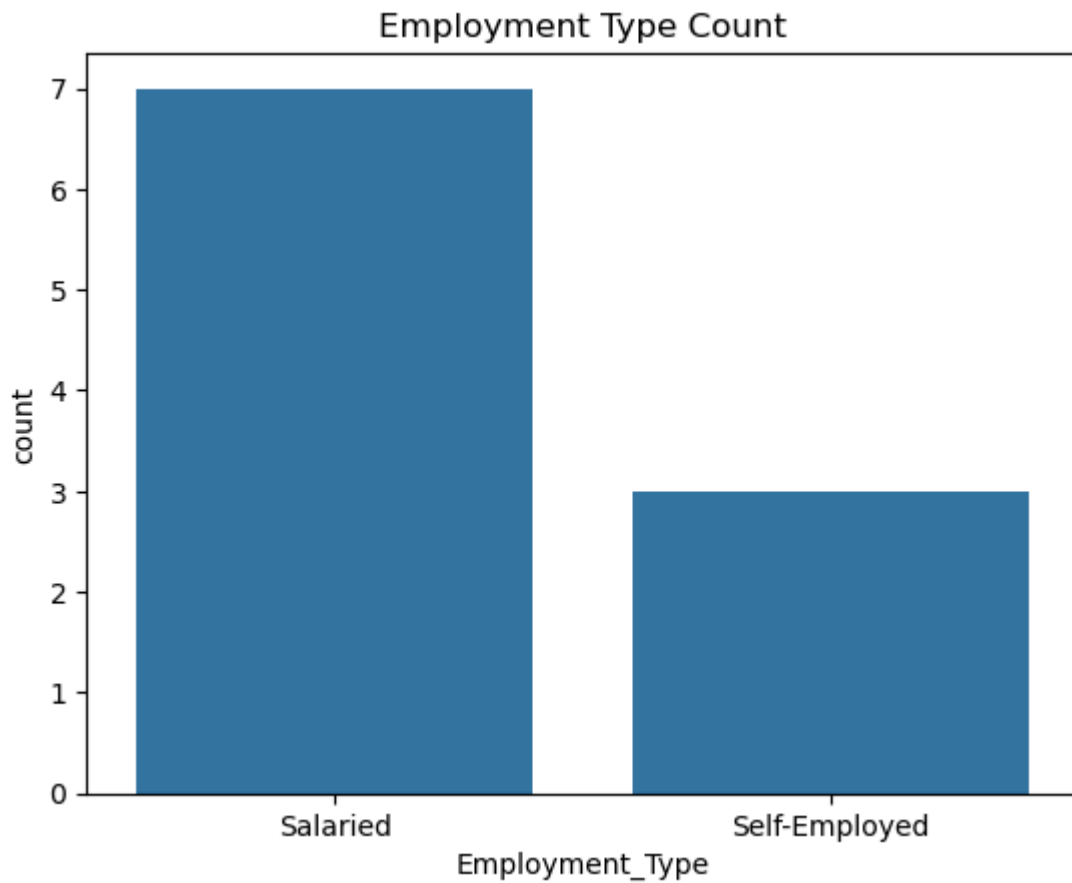


```
In [60]: credit_counts = Loan_Data["Credit_Score"].value_counts()
```

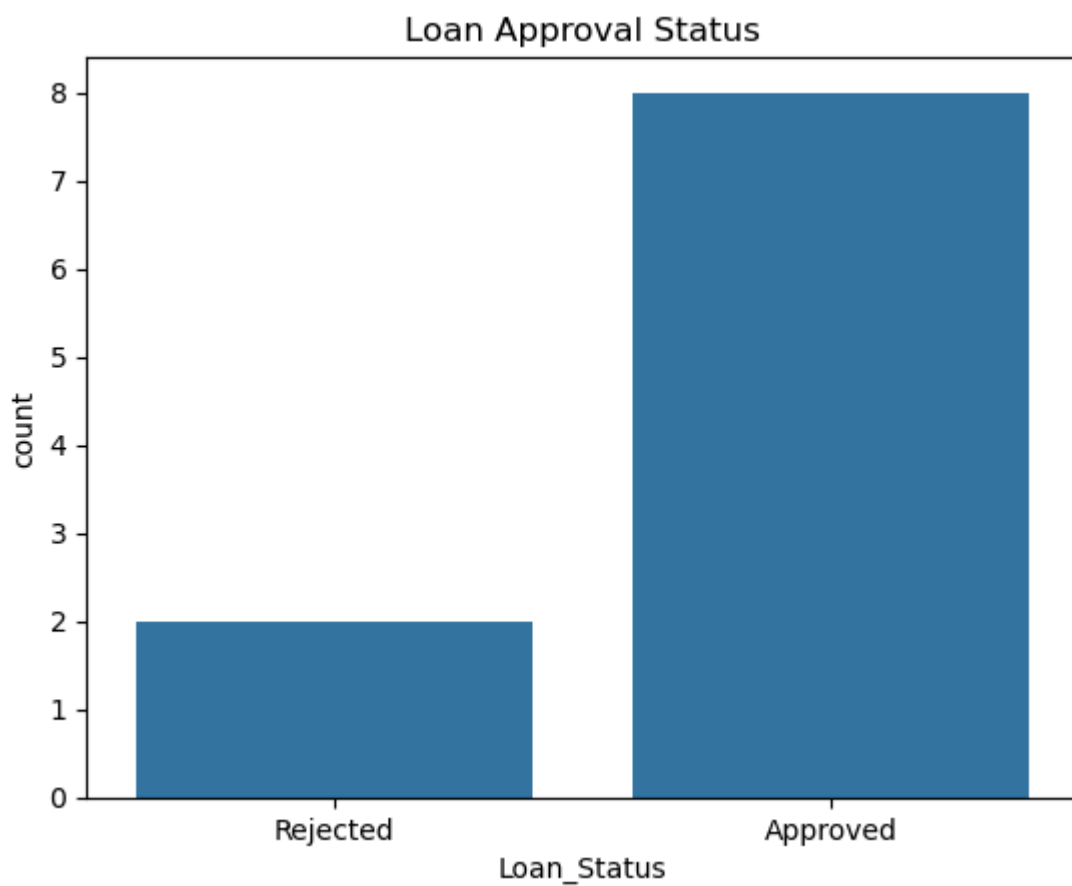
```
In [61]: # Insight: Bar chart
credit_counts.plot(kind="bar")
plt.title("Credit Score Distribution")
plt.xlabel("Credit Score")
plt.ylabel("Number of Applicants")
plt.show()
```



```
In [62]: # Insight: Employment Type Count
sns.countplot(x="Employment_Type", data=Loan_Data)
plt.title("Employment Type Count")
plt.show()
```

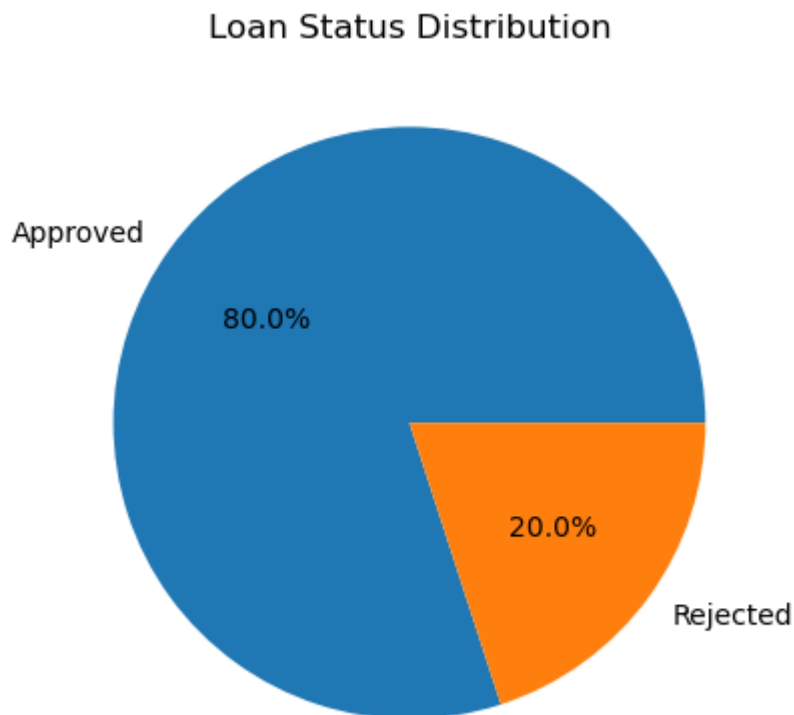


```
In [63]: # Insight: Loan Status Count
sns.countplot(x="Loan_Status", data=Loan_Data)
plt.title("Loan Approval Status")
plt.show()
```

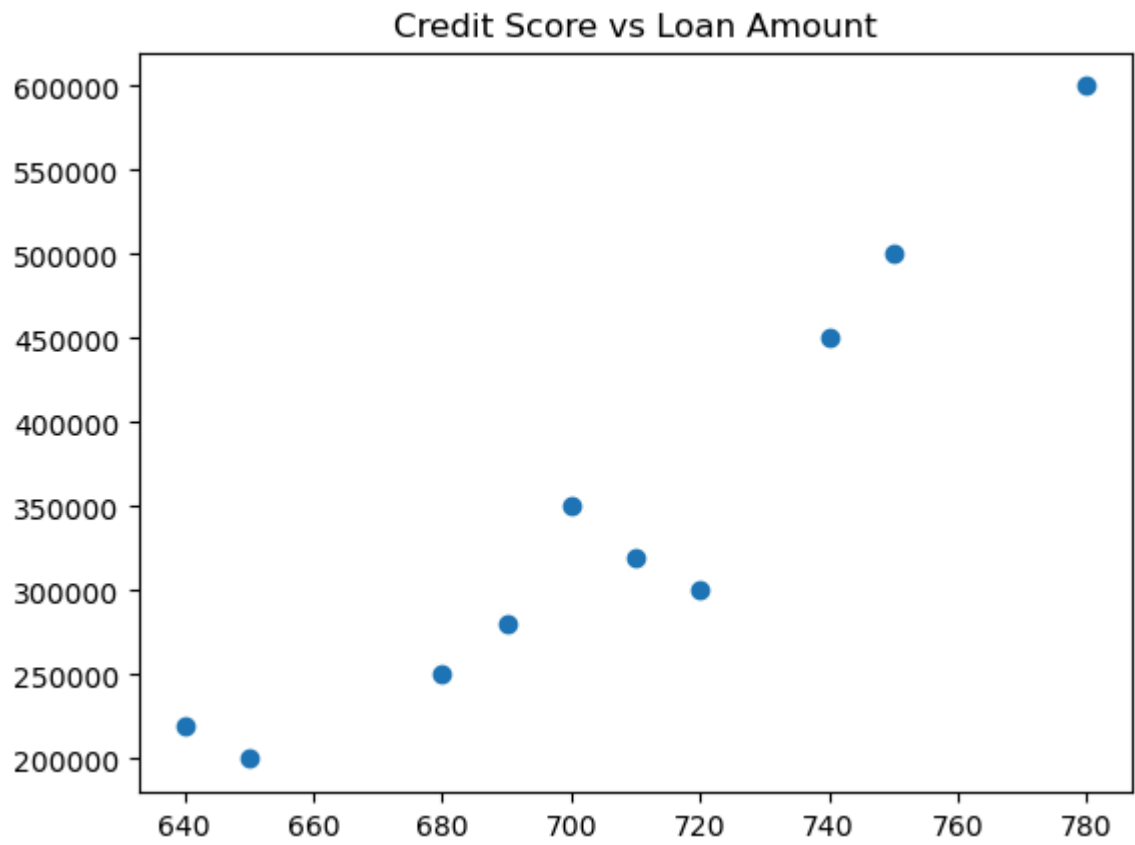


```
In [64]: loan_status_counts = Loan_Data["Loan_Status"].value_counts()
```

```
In [65]: # Insight: Pie chart
plt.pie(loan_status_counts, labels=loan_status_counts.index, autopct='%1.1f%%')
plt.title("Loan Status Distribution")
plt.show()
```



```
In [66]: # Insight: Scatter plot
plt.scatter(Loan_Data["Credit_Score"], Loan_Data["Loan_Amount"])
plt.title("Credit Score vs Loan Amount")
plt.show()
```



In []: